

Business Requirements

Employee Login & Logout Analysis

SQL Case Study | MySQL 8.0+

1. Business Objective

The organization wants to analyze employee login activity and session performance to understand user engagement, identify inactive users, monitor quarterly activity, and recognize highly engaged employees.

The analysis should help management answer questions around:

- Employee activity and inactivity
- Login and session trends
- Quarterly changes in usage
- Employee engagement patterns
- Daily session performance
- Consistently active users

2. Business Requirements

Requirement	Business Question	Expected Output	Purpose
BR-01: User Inactivity Analysis	Which users have not logged in during the past 5 months?	User ID	Identify potentially inactive users.
BR-02: Quarterly Activity Analysis	How many users and sessions are recorded in each quarter?	Quarter, User Count, Session Count	Monitor activity trends.
BR-03: Login Behavior Analysis	Which users logged in during January 2025 but not during November 2025?	User ID	Identify changes in individual engagement.
BR-04: Quarter-over-Quarter Analysis	How has the number of sessions changed compared with the previous quarter?	Quarter, Current Sessions, Previous Sessions, % Change	Track session growth or decline.
BR-05: Daily Session Performance	Which user generated the highest total session score for each day?	Date, User Name, Score	Identify daily top performers.
BR-06: Highly Engaged User Analysis	Which users logged in every day from their first login through the analysis date?	User ID, First Login, Date Difference, Login Days	Identify consistently active users.

3. Detailed Business Requirements

BR-01: Identify Inactive Users

Management needs to identify employees who have not logged in during the previous five months.

Business Purpose:

Identify potentially inactive users and support workforce engagement or account review activities.

Business Rule:

- Analysis date: 11 July 2025.
- Find users whose latest login occurred before the five-month cutoff.
- Return the USER_ID.

BR-02: Quarterly Login & Session Analysis

Management wants to understand employee activity across quarters.

Business Purpose:

Monitor how user participation and session volume change over time.

Required Metrics:

- Number of unique users
- Number of sessions
- First day of each quarter

Output: First Day of Quarter, User Count, Session Count.

Sorting: Newest quarter → oldest quarter.

BR-03: Compare Login Activity Across Months

Management wants to identify users who were active in January 2025 but not in November 2025.

Business Purpose:

Understand changes in individual user engagement over time and identify users whose activity declined or stopped.

Business Rule:

- User must have at least one login in January 2025.
- User must have no login in November 2025.

Output: USER_ID.

BR-04: Quarter-over-Quarter Session Growth

Management wants to measure how session activity changes from one quarter to the next.

Business Purpose:

Track whether overall session activity is increasing or decreasing over time.

Required Metrics:

- Current quarter session count
- Previous quarter session count
- Percentage change from the previous quarter

Output: First Day of Quarter, Session Count, Previous Session Count, Session Count Percentage Change.

Formula:

Session % Change = (Current Quarter Sessions – Previous Quarter Sessions) / Previous Quarter Sessions × 100

BR-05: Daily Top Session Performer

Management wants to identify the employee with the highest total session score for each day.

Business Purpose:

Identify the highest-performing users based on daily session activity.

Business Rule:

- Aggregate session scores by user and date.
- Identify the user with the highest total score for each day.

Output: Date, User Name, Total Session Score.

BR-06: Identify Highly Engaged Users

Management wants to identify users who have maintained continuous daily login activity since their first login.

Business Purpose:

Recognize highly consistent users and understand long-term engagement.

Analysis Date: 28 July 2025.

Business Rule:

- Determine each user's first login date.
- Calculate the number of days from the first login to the analysis date.
- Count the number of distinct days on which the user logged in.
- Identify users whose login-day count matches the number of days they were expected to be active.

Output: USER_ID, First Login, Date Difference, Count of Login Days.

4. Key Business Metrics

User Activity

- Active Users
- Inactive Users

- Login Days
- First Login Date

Session Activity

- Total Sessions
- Sessions per Quarter
- Sessions per User
- Session Score

Engagement

- Daily Active Users
- Consistently Active Users
- Users with declining activity

Trend Analysis

- Quarterly Session Change
- Quarter-over-Quarter % Change

5. Expected Business Outcome

1. Identify inactive employees.
2. Monitor quarterly user and session activity.
3. Detect changes in individual login behavior.
4. Measure session growth or decline over time.
5. Identify the highest daily session performers.
6. Recognize consistently engaged users.
7. Use login and session data to support workforce engagement decisions.

6. Case Study Summary

The Employee Login & Logout Analysis case study uses MySQL to analyze employee login behavior, session activity, engagement patterns, and quarterly trends. The analysis applies SQL techniques such as aggregations, joins, subqueries, CTEs, window functions, date functions, and comparative analysis to convert raw login data into actionable workforce insights.